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## **E-233 UNDERGROUND ELECTRICAL DUCTS**

### **1. SCOPE OF WORK**

This Technical Specification covers the design, supply, and construction requirements for underground electrical duct which the Contractor will install for efficient, economical, safe routing of cables or 500 kV Compressed Gas Insulated Bus (CGIB) in accordance with Part 4, Subsection 1.2, Summary of Work.

Through this specification the word “cable duct” are used to refer to concrete lined trenches.

### **2. CODES AND STANDARDS**

The Contractor shall comply with the requirements of Part 4, Subsection 2.5, Codes and Standards, and the following specific standards.

#### **2.1 National**

PUIL : Peraturan Umum Instalasi Listrik-Indonesia  
(General Regulation of Electrical Installation-Indonesia)  
SPLN : Standard Perusahaan Umum Listrik Negara

#### **2.2 International Electrotechnical Commission**

IEC 60529 : Degrees of protection provided by enclosures

### **3. DESIGN AND ENVIRONMENTAL CONSIDERATIONS**

The Contractor shall provide for the Owner’s review the preliminary drawings that he needs to use for detailed design of his cable duct or CGIB. These drawings shall provide preliminary information for the cable ducts routing.

The Contractor shall avoid all interferences with buried pipes, culverts, drains, foundations and other structures in his routing and shall provide adequate design clearances. The Contractor shall have his detailed cable duct design and installation drawings for Owner approval.

## **4. DETAILED REQUIREMENTS**

### **4.1 General**

The underground duct shall be designed for efficient and safe installation of the cables or CGIB. The Contractor shall submit the cable pulling calculations, the cable duct design and Construction drawings, showing all details, for the Owner's approval.

The duct shall be provided with drainage system in accordance with the Technical Specification S-001 and Bid Drawings 1222-00-S-07-UZF-001.

The grounding cables for the duct bank shall be at least 240 mm<sup>2</sup> lead covered copper.

All cables running in the cable duct shall be supported on plastic coated steel or fiberglass ladder type cable trays.

Cable duct shall continue inside buildings and the tray system interconnected to the building tray system as necessary for continued routing of the cables. Suitable access ladders or stairways shall be provided at both ends of the duct and at intermediate points as necessary for access.

### **4.2 Cable Tray and Conduit Installation**

Cable tray and conduit installation shall be in accordance with Technical Specification E-231 and E-232 respectively.

### **4.3 Reinforcing and Concrete**

Reinforcing is required in all cable ducts including covers.

Reinforcing bars should be placed with 50 mm clearance (minimum) to the outside of the concrete envelope and 50 mm clearance (minimum) to the outside of the conduits.

Concrete work shall be in accordance with Part 4, Technical Specification S-340.

Reinforcing steel and wire hair pins shall be set in the bottom slab.

**4.4           Excavation and Backfill**

Earthwork shall be in accordance with the Part4, Technical Specification S-300.

**5.             TESTS**

The cable tray and conduits shall meet the test requirements of their respective industry standards.

Concrete and concrete work shall meet the test requirements listed in Technical Specification S-340.

**6.             QUALITY ASSURANCE/QUALITY CONTROL**

Quality assurance/quality control for underground electrical ducts shall conform to the requirements of in Part .4, Subsection 2.7.

The quality assurance/quality control requirements of Technical Specifications S-300 and S-340 shall also apply.

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